Fiscal Year:	FY 2019	**Task Last Updated:**	FY 01/15/2019
PI Name:	Kernagis, Dawn Ph.D.		
Project Title:	Cervical Lymphatic Function Quantification and Associated Molecular Changes in Response to Simulated Microgravity		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32502-6008	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 HERO NNJ16ZSA001N-TRIRT. Appendix C: Translational Research Institute for Space Health (TRISH) Research Topics
Start Date:	10/01/2017	**End Date:**	03/31/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2020 per E. Urquieta/TRISH (Ed., 5/23/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clark, Jonathan M.D. (Florida Institute For Human And Machine Cognition Inc.) Crandall, David Ph.D. (Indiana University) Sevick, Eva Ph.D. (University of Texas Health Science Center)		
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Task Description:	An increased presence of white matter hyperintensities (WMH) has been previously described in astronauts; however, the underlying cause of this pathology is not clearly understood. Astronauts exposed to long-duration microgravity also develop varying degrees of visual change and signs of elevated intracranial pressure (ICP), which are thought to potentially be associated with the increased WMH seen in this population. Central nervous systems (CNS) changes associated with extreme environmental exposures, including increased WMH, have also been detected in high altitude aviators and professional SCUBA divers. There is growing evidence that a proportion of the cerebrospinal fluid drains from the cranium via the deep cervical lymphatic pathway. Recent literature has also revealed the existence of lymphatic vessels in the brain that serve to regulate fluid pressure and pathological clearance for surrounding tissue, and these vessels are connected to the peripheral lymphatics. Impairment of these drainage systems in other pathological models have demonstrated histological changes, including tau protein accumulation and white matter damage, in the brain. Changes in fluid dynamics with environmental exposures, such as microgravity or immersion, could lead to impairment of brain lymphatic drainage, resulting in accumulation fluid and elevated ICP, stasis of brain metabolic waste, and subsequent tissue damage and lesion development. At present, however, the effect of microgravity on deep cervical lymphatic function has not yet been characterized. Impact: Understanding the effect of microgravity on cervical lymphatic function and outflow from the brain will provide insight into factors and potential mitigation strategies for dangerous training or mission-limiting physiological conditions, such as increased ICP and development of WMH, which could detrimentally impact astronaut performance both acutely and long-term. In addition, further refinement of lymphatic function monitoring technology could lead to its utilization in space as a diagnostic tool for crew members experiencing CNS symptoms. The proposed study will address the following Specific Aims: SA1: Refine the current near-infrared fluorescence lymphatic imaging (NIRFLI) system underwater lymphatic imaging. SA2: Characterize and compare the effect of deep cervical lymphatic function in humans in response to two models of simulated microgravity: head-out immersion (HOI) and head-down tilt (HDT). SA3: Develop a visual quantification software program for objectively measuring lymphatic function as captured by the device. SA4: Evaluate targeted molecular biomarkers associated with impaired immune function and lymphatic clearance in response to the HOI and HDT models of simulated microgravity. Approach: The proposed study will refine existing lymphatic imaging technology while addressing the hypothesis that simulated microgravity (HDT, HOI) in humans impairs lymphatic transit speed in the head/ neck lymphatics as compared to baseline due to the associated increase in hydrostatic pressure on the lymphatic system while increasing lymphatic transit in a chest- and head-ward direction. Study participants will undergo dry baseline measurements of cervical lymphatic function using a novel, portable, low footprint method for assessing lymphatic function, followed by measurements conducted during and following simulated microgravity exposures. Each subject will undergo both HDT and HOI, separated by at least 48 hours. Venous blood samples will be collected for assessment of targeted biomarkers prior to baseline measurements and at the conclusion of each exposure. In parallel with study data collection, visual analysis and machine learning algorithms will be applied to develop a software program that automatically quantifies lymphatic function in conjunction with the measuring device.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this study are pertinent to NASA's investigation into the incidence of Spaceflight-Associated Neuro-ocular Syndrome and the impacts of fluid shift and immune dysfunction in microgravity. This application will further refine and develop high-impact, low-footprint technology to monitor lymphatic function in humans that could potentially be used as a diagnostic tool during long-duration spaceflight.
Task Progress:	Original Aims and Objectives: Understanding the effect of microgravity on cervical lymphatic function could provide insight into factors and potential mitigation strategies for dangerous training or mission-limiting physiological conditions, which could detrimentally impact astronaut performance both acutely and long-term. Further development of lymphatic function monitoring technology could also be utilized in space as a diagnostic tool for crew members. The proposed study will address the following Specific Aims: Specific Aim 1: Characterize the response of lymphatic function humans in response to the head-out immersion and head-down tilt models of simulated microgravity. We will perform baseline imaging of subjects (n = 16) in a dry, seated position. Subjects will then undergo either a dry head-down tilt exposure or a head-out immersion exposure and imaging of lymphatic transit time will be conducted. Each subject will undergo each experimental exposure once, with a minimum of 24 hours between exposures. Specific Aim 2: Develop a visual quantification software program for objectively measuring lymphatic function based on images captured by the device. Specific Aim 3: Evaluate targeted peripheral biomarkers to begin characterization of impaired brain lymphatic clearance at the molecular level. Venous blood samples will be collected before and after each experimental exposure for targeted biomarker evaluation. Key Findings to Date: The subcontracted study team at Indiana University (Principal Investigator David Crandall) is leading the software development, and initial data assessment and visual quantification is underway utilizing anonymized, previously collected data for development and initial software validation. The subcontracted team at University of Texas Health Science Center (Principal Investigator Eva Sevick) is leading the underwater lymphatic imaging device development. The team conducted initial studies to assess the ability of the camera system to capture an in-water signal through a glass window, which was successful. Human protocol optimization and testing at the Florida Institute For Human And

	Machine Cognition Inc. (IHMC) are underway. Impact of key findings on hypotheses, technology requirements, objectives, and specific aims of the original proposal: The testing of the lymphatic imaging device demonstrated feasibility of image capture through a glass barrier and through water. Visual quantification of the device data will allow for objective measurement of lymphatic function, which was a primary goal of the proposed study. Pilot feasibility testing of the human study protocol has been completed and prepared for testing with the lymphatic device. Results: Results to date demonstrate feasibility of the proposed study objectives.
Bibliography Type:	Description: (Last Updated: 01/09/2019)
Awards	Kernagis D. "Fellow, National Explorers Club, April 2018." Apr-2018
Awards	Kernagis D. "Young Scientist Award, Undersea and Hyperbaric Medical Society, May 2018." May-2018